

MATH 2211, SPRING 2026: SOLUTIONS TO HOMEWORK 10

- (1) Suppose that $T : V \rightarrow V$ is a linear transformation and that $\mathcal{B}_1$ and $\mathcal{B}_2$ are bases for V . Show that $[T]_{\mathcal{B}_1}$ and $[T]_{\mathcal{B}_2}$ have the same minimal polynomial.

Solution. Suppose that $\mathcal{B}_1 = (v_1, \dots, v_n)$ and $\mathcal{B}_2 = (w_1, \dots, w_n)$. Let C be the change of basis matrix $[c_{i,j}]$ where

$$w_j = \sum_{i=1}^n c_{i,j} v_i.$$

Then by what we saw in Lecture 23, we have

$$[T]_{\mathcal{B}_1} = C[T]_{\mathcal{B}_2} C^{-1}.$$

We can now apply Observation 3 in Lecture 31 to conclude.

- (2) What are the minimal and characteristic polynomials of the matrix

$$A = \begin{bmatrix} 2 & 1 & 0 & 0 \\ 0 & 2 & 0 & 0 \\ 0 & 0 & 2 & 0 \\ 0 & 0 & 0 & 3 \end{bmatrix}$$

?

Solution. Since A is upper triangular, we can read off the characteristic polynomial and the eigenvalues from the diagonal entries (see Observation 5 in Lecture 27). This gives us

$$\text{ch}_A(x) = (x - 2)^3(x - 3).$$

To find the minimal polynomial, we need to know the minimal power m of $x - 2$ such that $(A - 2I_4)^m(A - 3I_4) = 0$. For this, we look at

$$A - 2I_4 = \begin{bmatrix} 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

This corresponds to the linear transformation $e_1 \mapsto 0, e_2 \mapsto e_1, e_3 \mapsto 0, e_4 \mapsto e_4$. Doing it twice would give us $e_1 \mapsto 0, e_2 \mapsto 0, e_3 \mapsto 0, e_4 \mapsto e_4$. This tells us that $(A - 2I_4)^2$ kills e_1, e_2, e_3 . Since $A - 3I_4$ kills e_4 , we see that

$$(A - 2I_4)^2(A - 3I_4) = 0.$$

On the other hand, $(A - 2I_4)(A - 3I_4) \neq 0$, since we have

$$(A - 2I_4)(A - 3I_4)e_2 = (A - 2I_4)(Ae_2 - 3e_2) = (A - 2I_4)(e_1 + 2e_2 - 3e_2) = (A - 2I_4)(e_1 - e_2) = -e_1.$$

This shows that the minimal polynomial is $(x - 2)^2(x - 3)$.

- (3) Suppose that $A \in M_{n \times n}(F)$ is such that $\min_A(x)$ is a multiple of $(x - \lambda)^2$ for some $\lambda \in F$. Show that A cannot be diagonalizable.

Solution. You can use the criterion from the Proposition in Lecture 31: In particular, you can use the implication (1) $\Rightarrow$ (3) shown on page 4 there.

Definition 1. If $A \in M_{n \times n}(\mathbb{C})$ and $\lambda \in \mathbb{C}$, then the **generalized λ -eigenspace** for A is

$$E_{\lambda}^{\text{gen}} = \{v \in \mathbb{C}^n : (A - \lambda I_n)^m v = 0 \text{ for some } m \geq 1\}.$$

A non-zero vector in E_{λ}^{gen} is called a **generalized eigenvector** with eigenvalue λ .

(4) Show that the following are equivalent for an eigenvalue $\lambda \in \mathbb{C}$ for A :

- (a) $E_{\lambda}^{\text{gen}} = E_{\lambda}$.
- (b) The minimal polynomial $\min_A(x)$ is a multiple of $x - \lambda$ but not of $(x - \lambda)^2$.

Solution. The first condition says that any vector killed by $(A - \lambda I_n)^m$ for $m \geq 1$ is actually killed by $A - \lambda I_n$. The second condition says that the minimal polynomial is of the form

$$\min_A(x) = (x - \lambda)q(x),$$

where $q(\lambda) \neq 0$.

Let us show (a) $\Rightarrow$ (b). Note that, since λ is an eigenvalue, we already know that $\min_A(\lambda) = 0$ (why?), which means that $\min_A(x) = (x - \lambda)q(x)$ for some polynomial $q(x) \in \mathbb{C}[x]$. We want to know that $q(\lambda) \neq 0$. In fact, we will prove the contrapositive: If $q(\lambda) = 0$, then $E_{\lambda} \neq E_{\lambda}^{\text{gen}}$. Indeed, in this case we would have $\min_A(x) = (x - \lambda)^2 q_1(x)$, where $q(x) = (x - \lambda)q_1(x)$. This implies two things:

- $(A - \lambda I_n)^2 q_1(A) = \min_A(A) = 0$;
- There is a vector $v \in \mathbb{C}^n$ such that $(A - \lambda I_n)q_1(A)v \neq 0$: If this were not the case, then we would have $(A - \lambda I_n)q_1(A) = 0$, which would contradict the minimality of $\min_A(x)$.

Consider the vector $w = q_1(A)v$: this is non-zero (why?) and we have $(A - \lambda I_n)^2 w = 0$ (why?), but $(A - \lambda I_n)w \neq 0$ (why?). In other words, $w \in E_{\lambda}^{\text{gen}}$, but $w \notin E_{\lambda}$. This shows the contrapositive implication.

Now let us show (b) $\Rightarrow$ (a): Again, it's easier to show the contrapositive. If $E_{\lambda} \neq E_{\lambda}^{\text{gen}}$, there exists w such that $(A - \lambda I_n)^2 w = 0$ but $(A - \lambda I_n)w \neq 0$ (why?). Here is the key observation:

Observation 1. If $w \in E_{\lambda}^{\text{gen}}$ is a generalized eigenvector, and $\mu \neq \lambda$, then $(A - \mu I_n)w \neq 0$.

Proof. Indeed, if this were not the case, then we would have $Aw = \mu \cdot w$, which would imply that $(A - \lambda I_n)w = (\mu - \lambda)w$, and hence $(A - \lambda I_n)^m w = (\mu - \lambda)^m w \neq 0$. This means that w could never be a generalized eigenvector for A with eigenvalue λ . $\square$

This observation tells us that, if $h(\lambda) \neq 0$, then $h(A)w \neq 0$. It also tells us that, $h(A)(A - \lambda I_n)w \neq 0$ (why?). Therefore, $(A - \lambda I_n)h(A) \neq 0$, and so the minimal polynomial has to be divisible by at least $(x - \lambda)^2$.

Assume the following result:

Theorem 1. F^n admits a basis consisting of generalized eigenvectors for A .

(5) Using the above theorem, show that $\dim E_{\lambda}^{\text{gen}} = m(\lambda)$, where $m(\lambda)$ is the algebraic multiplicity, and conclude that A is diagonalizable whenever $\min_A(x)$ has no repeated linear factors.

Solution. Suppose that the distinct eigenvalues are $\lambda_1, \dots, \lambda_r$ and that we have a basis of generalized eigenvectors

$$\{v_{1,1}, \dots, v_{1,m_1}, \dots, v_{r,1}, \dots, v_{r,m_r}\}$$

where $v_{i,j}$ is a generalized eigenvector with eigenvalue λ_i . We want to show that $m_i = m(\lambda_i)$. It is actually enough to show that $m_i \leq m(\lambda_i)$. This is because we have $n = \sum_{i=1}^r m_i$ and $n = \sum_{i=1}^r m(\lambda_i)$ (why?).

To show this inequality, it is enough to show $\dim E_{\lambda}^{\text{gen}} \leq m(\lambda)$ for every λ . For this, note that multiplication by A restricts to a linear transformation $T : E_{\lambda}^{\text{gen}} \rightarrow E_{\lambda}^{\text{gen}}$. This amounts to knowing that if w is a generalized eigenvector with eigenvalue λ , then so is Aw : This is clear, since $(A - \lambda I_n)^m Aw = A(A - \lambda I_n)^m w = 0$ (we are using the fact that A and $A - \lambda I_n$ commute with each other). Moreover, by Observation 1 above, this linear transformation can only have λ as an eigenvalue (why?). Therefore, its characteristic polynomial must be $(x - \lambda)^{\dim E_{\lambda}^{\text{gen}}}$. This cannot have more factors of $x - \lambda$ than what we have in $\text{ch}_A(x)$ (why?), which tells us that $\dim E_{\lambda}^{\text{gen}} \leq m(\lambda)$.

Now, for the last part: By the previous problem, we know that, since $\min_A(x)$ has no repeated factors, for every eigenvalue λ , we have $E_\lambda = E_\lambda^{\text{gen}}$. In other words, every generalized eigenvector is actually an eigenvector. Since we have a basis of generalized eigenvectors, we must actually have a basis of eigenvectors: That is, A is diagonalizable.

- (6) Use least squares approximation to find the cubic function $f(x) = a_3x^3 + a_2x^2 + a_1x + a_0 \in \text{Poly}(\mathbb{R})$ that best minimizes the quantity

$$f(-2)^2 + (f(-1) + 1)^2 + f(0)^2 + (f(1) - 2)^2 + (f(2) - 1)^2.$$

Solution. Looking at Lecture 32, we see that we are trying to find a least squares approximate solution to the equation

$$\begin{bmatrix} -1 & 1 & -1 & 1 \\ 0 & 0 & 0 & 1 \\ 1 & 1 & 1 & 1 \\ 8 & 4 & 2 & 1 \\ -8 & 4 & -2 & 1 \end{bmatrix} x = v = \begin{bmatrix} -1 \\ 0 \\ 2 \\ 1 \\ 0 \end{bmatrix}$$

If $A = \begin{bmatrix} -1 & 1 & -1 & 1 \\ 0 & 0 & 0 & 1 \\ 1 & 1 & 1 & 1 \\ 8 & 4 & 2 & 1 \\ -8 & 4 & -2 & 1 \end{bmatrix}$, then we have

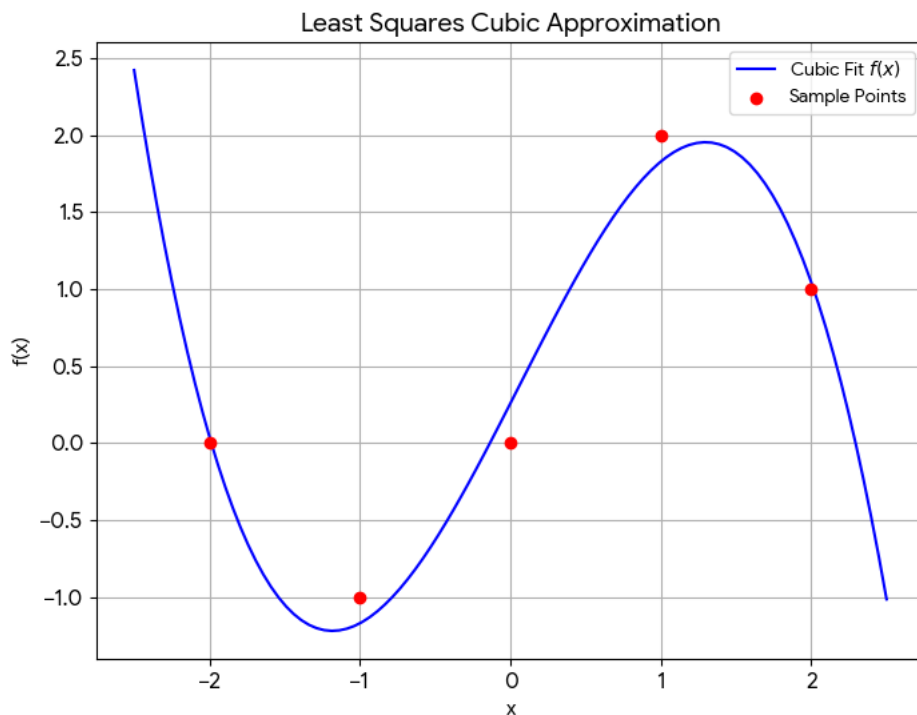
$$A^T A = \begin{bmatrix} 130 & 0 & 34 & 0 \\ 0 & 34 & 0 & 10 \\ 34 & 0 & 10 & 0 \\ 0 & 10 & 0 & 5 \end{bmatrix}, A^T v = \begin{bmatrix} 11 \\ 5 \\ 5 \\ 2 \end{bmatrix}$$

We have to now solve for $A^T A x = A^T v$. The solution turns out to be unique:

$$a_3 = -\frac{5}{12}, \quad a_2 = \frac{1}{14}, \quad a_1 = \frac{23}{12}, \quad a_0 = \frac{9}{35}.$$

This gives us the cubic polynomial

$$f(x) = -\frac{5}{12}x^3 + \frac{1}{14}x^2 + \frac{23}{12}x + \frac{9}{35}.$$



(7) Consider

$$A = \begin{bmatrix} 1 & 1 & 0 \\ 1 & 0 & 1 \\ 0 & 1 & 1 \end{bmatrix} \in M_{3 \times 3}(\mathbb{R})$$

Use the Gram-Schmidt process to write this as a product $A = QR$, where Q is orthonormal and where R is upper triangular.

Solution. We have $v_1 = \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix}$, $v_2 = \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix}$, $v_3 = \begin{bmatrix} 0 \\ 1 \\ 1 \end{bmatrix}$. It's not hard to see that these vectors form a basis.

Now we run Gram-Schmidt:

(a) $u_1 = \frac{v_1}{\|v_1\|} = \begin{bmatrix} 1/\sqrt{2} \\ 1/\sqrt{2} \\ 0 \end{bmatrix}$. Note that we have $v_1 = \sqrt{2}u_1$.

(b) $v'_2 = v_2 - (v_2 \cdot u_1)u_1$. This gives us

$$v'_2 = \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix} - \frac{1}{\sqrt{2}} \begin{bmatrix} 1/\sqrt{2} \\ 1/\sqrt{2} \\ 0 \end{bmatrix} = \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix} - \begin{bmatrix} 1/2 \\ 1/2 \\ 0 \end{bmatrix} = \begin{bmatrix} 1/2 \\ -1/2 \\ 1 \end{bmatrix}$$

Note that $v_2 = \frac{1}{\sqrt{2}}u_1 + v'_2$.

(c) $u_2 = \frac{v'_2}{\|v'_2\|} = \frac{\sqrt{2}}{\sqrt{3}}v'_2 = \begin{bmatrix} 1/\sqrt{6} \\ -1/\sqrt{6} \\ \sqrt{2}/\sqrt{3} \end{bmatrix}$. Note that $v_2 = \frac{1}{\sqrt{2}}u_1 + \frac{\sqrt{3}}{\sqrt{2}}u_2$.

(d) $v'_3 = v_3 - (v_3 \cdot u_1)u_1 - (v_3 \cdot u_2)u_2$. This gives us

$$\begin{aligned} v'_3 &= \begin{bmatrix} 0 \\ 1 \\ 1 \end{bmatrix} - \frac{1}{\sqrt{2}}u_1 - \frac{1}{\sqrt{6}}u_2 \\ &= \begin{bmatrix} 0 \\ 1 \\ 1 \end{bmatrix} - \begin{bmatrix} 1/2 \\ 1/2 \\ 0 \end{bmatrix} - \begin{bmatrix} 1/6 \\ -1/6 \\ 1/3 \end{bmatrix} \\ &= \begin{bmatrix} -2/3 \\ 2/3 \\ 2/3 \end{bmatrix} \end{aligned}$$

Note that $v_3 = \frac{1}{\sqrt{2}}u_1 + \frac{1}{\sqrt{6}}u_2 + v'_3$.

(e) $u_3 = \frac{v'_3}{\|v'_3\|} = \frac{\sqrt{3}}{\sqrt{4}}v'_3 = \begin{bmatrix} -1/\sqrt{3} \\ 1/\sqrt{3} \\ 1/\sqrt{3} \end{bmatrix}$. Note that $v_3 = \frac{1}{\sqrt{2}}u_1 + \frac{1}{\sqrt{6}}u_2 + \frac{2}{\sqrt{3}}u_3$.

Putting everything together, we find

$$Q = \begin{bmatrix} 1/\sqrt{2} & 1/\sqrt{6} & -1/\sqrt{3} \\ 1/\sqrt{2} & -1/\sqrt{6} & 1/\sqrt{3} \\ 0 & \sqrt{2/3} & 1/\sqrt{3} \end{bmatrix}; R = \begin{bmatrix} \sqrt{2} & 1/\sqrt{2} & 1/\sqrt{2} \\ 0 & \sqrt{3/2} & 1/\sqrt{6} \\ 0 & 0 & 2/\sqrt{3} \end{bmatrix}$$

(8) Suppose a bacterial population $P(t)$ (in thousands) is observed at time t (in hours). The growth follows an exponential model:

$$P(t) = Ae^{kt}$$

where A is the initial population and k is the growth rate. Given the following data, find the best fit for A and k using the method of least squares.

Time t (hr)	1	2	3	4
Population P	2.1	4.5	9.2	18.8

Hint: Linearize the problem by taking logarithms!

Solution. As the hint says, we look at the problem

$$\ln(P(t)) = \ln(A) + kt$$

instead. If $C = \ln(A)$, then we have to solve for k and C that best approximates the given values. This amounts to finding a least squares solution to the equation

$$Bx = \begin{bmatrix} 1 & 1 \\ 1 & 2 \\ 1 & 3 \\ 1 & 4 \end{bmatrix} x = \begin{bmatrix} \ln(2.1) \\ \ln(4.5) \\ \ln(9.2) \\ \ln(18.8) \end{bmatrix} = v$$

We have

$$B^T B = \begin{bmatrix} 4 & 10 \\ 10 & 30 \end{bmatrix}; B^T v = \begin{bmatrix} \ln(2.1 * 4.5 * 9.2 * 18.8) \\ \ln(2.1 * (4.5)^2 * (9.2)^3 * (18.8)^4) \end{bmatrix} \simeq \begin{bmatrix} 7.3991 \\ 22.1431 \end{bmatrix}$$

The solution to $B^T Bx = B^T v$ is now given up to four digit precision by

$$x = \begin{bmatrix} C \\ k \end{bmatrix} = \begin{bmatrix} 0.0270 \\ 0.7291 \end{bmatrix}$$

So with $A = e^C \simeq 1.0274$, we see that a good approximation to the original problem is given by $P(t) = 1.0274e^{0.7291t}$.